

Principles of Computer Security, Sixth Edition

Chapter 17 Solutions

Key Terms Quiz

1. unsolicited commercial e-mail
2. DomainKeys Identified mail (DKIM)
3. botnet
4. Multi-Purpose Internet Mail Extensions (MIME)
5. Real-time Blackhole List (RBL)
6. Open relay
7. Sender Policy Framework
8. plug-in, or browser helper object, or browser extension
9. Instant messaging (IM)
10. Simple Mail Transfer Protocol (SMTP)

Multiple-Choice Quiz

1. **A.** Zombie botnets send the majority of spam; using them is one of the ways that criminals can profit from Internet fraud.
2. **B.** Cookies with the **[secure]** tag are passed by browsers only over HTTPS connections.
3. **D.** Open e-mail relays allow anyone to send e-mail through the server.
4. **C.** JavaScript security is ultimately the responsibility of the end user, and the options exist in browsers to select various security levels or even disable it altogether.
5. **B.** Code signing includes data integrity checking through a hash value.
6. **A.** File sharing through instant messaging can bypass server-based virus protections.
7. **B.** The public key is used to encrypt the symmetric key, which is then used to encrypt the message contents, because encrypting the entire message would take too much processing power.

8. **C.** HTML e-mail can carry embedded instructions to download or run scripts that can be launched from the preview pane in some e-mail programs, without requiring that the user actively launch the attached program.
9. **C.** IM protocols require the traffic to travel to the hosting server, so two users within an organization are sending the traffic to an outside server and back when communicating via IM.
10. **D.** TLS provides all three types of service.

Essay Quiz

1. A successful spam policy should be built on strong upstream filtering, integrating some form of DNS-based source filtering. Strong keyword filtering also helps once the spam has reached the organization. Greylisting or SPF may also be used to implement spam policy.
2. The memo should talk about how malware can be transmitted through e-mail and IM programs, as well as how malware can give full control of your computer to an attacker, how all your personal or professional data could be stolen by the attacker, and how an attacker can use your computer to attack others on the network. The memo should also address methods of virus detection, such as strange reboots or a cursor moving on its own, as well as whom to contact when the behavior is detected. It should also speak about how to use the desktop antivirus system.